

EE5203 Microprocessors

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This is the 2026-2027 working site. Lecture 1 is the first renovated reference package. Lecture 2 onward remains migrated source material until it is revised against the approved Lecture 1 pattern.

Design baseline

- [Course structure and instructional sequence](#)
- [Term delivery map](#)
- [Academic references and further reading](#)
- [From C to Silicon — toolchain, memory, and boot companion](#)

The design baseline establishes a four-lecture Embedded C Foundation, a maximum of 35 core theory slides, fading laboratory scaffolding, register literacy with HAL fluency, and an early milestone-based project runway.

Where Things Go

- `lectures/` contains stable lecture-sequence sources (L01, L02, ...).
- `exams/` contains exam sources and print material.
- `private/` contains solutions, submissions, grades, and sensitive material.
- `admin/` contains syllabus, schedule, class-list, CATS/Orion, and course-file records.
- `_site/` contains rendered output after `quarto render`.

Rendered material should be found in `_site/`, not `dist/`.

Lecture numbers describe instructional order, not calendar weeks. The term-specific [delivery map](#) records when each lecture or exam occurs.

Lecture 1 - Embedded Systems, the C Toolchain, and Debugging

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)

- [Embedded C readiness diagnostic](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide](#)
- [Lab reference](#)

Worked solutions and instructor guidance are kept under `private/` until their release points.

Approved 2026-2027 sequence

Lectures 1-4 form a cumulative Embedded C foundation. Each laboratory begins from the previous week's working project.

Lecture 2 — Values, Decisions, Loops, and Arrays

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide](#)

Lecture 3 — Functions, Arrays, Strings, and Multiple C Files

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide](#)
- [Starter files](#)

Lecture 4 — Pointers, Structures, Bits, Registers, and GPIO

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide — Bare-Metal GPIO](#)
- [Homework — External GPIO Control](#)

Lecture 5 — GPIO and Hardware Abstraction

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide — GPIO with the HAL](#)

Lecture 6 — Interrupts and the NVIC

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide — Button Interrupts \(EXTI\)](#)
- [Homework — HAL Interrupts](#)

Lecture 7 — Timers (TIM)

- [Theory slides](#)
- [Lab guide — Timer Interrupts](#)

Lecture 8 — Pulse-Width Modulation

- [Theory slides](#)
- [Lab guide — Fading LEDs and Controlling Servos](#)

Lecture 9 — Input Capture

- [Theory slides](#)
- [Lab guide — Measure Your Own PWM](#)

Lecture 10 — Analog-to-Digital Conversion (ADC)

- [Theory slides](#)
- [Lab guide — Reading the Analog World](#)

Lecture 11 — The Bridge to Assembly

- [Theory slides](#)
- [Lab guide — The Disassembly View](#)

Lecture 12 — Synchronous Serial: I2C and SPI

- [Theory slides](#)

- [Lab slides — Analog Sensors and LCDs](#)
- [Lab slides \(potentiometer version\)](#)

Lecture 13 — DMA, Displays, and Motors

- [Theory slides](#)
- [Lab slides — TFT Display and SPI with DMA](#)
- [DMA deck extract](#)

Lecture 14 — Final Project Presentations and Course Wrap-up

- [Theory slides](#)
- [Lab slides — Final Project Development and Demonstrations](#)